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Securing Cryptocurrency Exchanges: Optimal Policy Design

Bachelor's Thesis

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December 21, 2025

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STUDENT DECLARATION

I, *Sen Yagiz Tuna*, the undersigned, hereby declare that the present Bachelor's Thesis work has been prepared by myself and without any unauthorized help or assistance. Only the specified sources (references, tools, etc.) were used. All parts taken from other sources word by word, or after rephrasing but with identical meaning, were unambiguously identified with explicit reference to the sources utilized. In the Appendix, I have clearly indicated whether I used artificial intelligence tools in preparing the thesis; if so, I have specified the manner and extent of their use in the table. I acknowledge that I bear full responsibility for any content generated by artificial intelligence – regardless of the extent of its use.

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Budapest, December 21, 2025

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Kivonat

A kriptovaluta-ökoszisztéma gyors bővülése a digitális eszközöket egy szűk technológiai kísérletből globálisan jelentős befektetési eszközosztállyá alakította. Minden egyes piaci ciklussal mind a lakossági, mind az intézményi részvétel számottevően növekedett, aminek következtében jelentős tőkemennyiség áramlott a kriptovaluta-tőzsdékre, amelyek mára a digitális eszközgazdaság alapvető infrastruktúráját alkotják. A korábbi tőzsdei összeomlások rávilágítottak a nem megfelelő vállalatirányítás, a gyenge működési kontrollok, az elégtelen átláthatóság és a hiányos kockázatkezelés súlyos következményeire, amelyek gyakran nagymértékű befektetői veszteségeket és rendszerszintű piaci zavarokat okoztak. Válaszul a szabályozó hatóságok olyan koherens keretrendszerek kialakítására törekedtek, amelyek védik a befektetőket, erősítik a piaci integritást, és ösztönzik a tőzsdéket a robusztus biztonsági és megfelelési standardok alkalmazására.

Ez a dolgozat játékelméleti megközelítést alkalmaz a szabályozási korlátok között működő kriptovaluta-tőzsdék stratégiai viselkedésének elemzésére, különös tekintettel az Európai Unió kriptoeszköz-piacokra vonatkozó rendeletére (Markets in Crypto-Assets, MiCA). A szabályozási követelmények stratégiai szakpolitikai döntésekként kerülnek modellezésre, ami lehetővé teszi annak vizsgálatát, hogy a különböző megfelelési stratégiák miként befolyásolják a tőzsdék teljesítményét, a piaci stabilitást és az össztársadalmi jólétet. A modell egy ismétlődő, több szereplős játékot ír le, amelyben a tőzsdék hosszú távon lépnek kölcsönhatásba egymással, és a szimulációk segítségével kerülnek értékelésre az egyensúlyi kimenetek az alternatív stratégiai profilok mellett.

A szimuláció eredményei azt mutatják, hogy a MiCA szakpolitikai követelményeinek teljes körű, összehangolt alkalmazása egyszerre alkot Nash-egyensúlyt és társadalmi optimumot. Amennyiben valamennyi tőzsde maradéktalanul végrehajtja a teljes szabályozási csomagot, egyik szereplőnek sem áll érdekében az eltérés, minden résztvevő pozitív kifizetést ér el, és az össztársadalmi jólét maximalizálódik. Ezzel szemben a modell azonosít egy olyan stabil egyensúlyt is, amelyet a költségminimalizáló magatartás jellemez, és amely minden tőzsde számára negatív kifizetéseket eredményez, jól szemléltetve a rövid távú ösztönzők által vezérelt koordinációs kudarcot. Ezek az eredmények azt igazolják, hogy az erőteljes szabályozási megfelelés koordinációs mechanizmusként működhet, amely összhangba hozza az egyéni profitorientált döntéseket a hosszú távú piaci stabilitással és a kollektív jóléttel.

A dolgozat fő hozzájárulása abban rejlik, hogy formális, stratégiai keretrendszert kínál a kriptoeszköz-szabályozás elemzésére mint ösztönzőrendszerre, nem pedig pusztán exogén korlátozásra. A szabályozási tervezés és az egyensúlyi kimenetek összekapcsolásával az elemzés új betekintést nyújt abba, miként járulhat hozzá az összehangolt megfelelés a kriptovaluta-piacok ellenálló képességének, bizalmának és fenntarthatóságának erősítéséhez. Az eredmények releváns következtetésekkel szolgálnak mind a szabályozási rezsimeket kialakító szakpolitikai döntéshozók, mind pedig azon tőzsdék számára, amelyek olyan stratégiákat keresnek, amelyek egyensúlyt teremtenek a jövedelmezőség és a hosszú távú piaci hitelesség között.

Abstract

The rapid expansion of the cryptocurrency ecosystem has transformed digital assets from a niche technological experiment into a globally significant investment class. With each market cycle, both retail and institutional participation has increased substantially, bringing large volumes of capital onto cryptocurrency exchanges that now serve as the core infrastructure of the digital asset economy. Historical exchange failures have demonstrated the severe consequences of inadequate governance, weak operational controls, insufficient transparency, and poor risk management, often resulting in large-scale investor losses and systemic market disruptions. In response, regulatory authorities have sought to develop coherent frameworks that protect investors, enhance market integrity, and incentivise exchanges to adopt robust security and compliance standards.

This thesis applies a game-theoretical approach to analyse the strategic behaviour of cryptocurrency exchanges operating under regulatory constraints, focusing on the European Union's Markets in Crypto-Assets (MiCA) framework. Regulatory requirements are modelled as strategic policy choices, allowing the examination of how different compliance strategies affect exchange performance, market stability, and aggregate welfare. A repeated multi-player game is constructed in which exchanges interact over long-term, and simulations are used to evaluate equilibrium outcomes across alternative strategy profiles.

The results of the simulation reveal that coordinated adoption of the full set of MiCA policy requirements constitutes both a Nash equilibrium and the social optimum. When all exchanges implement the complete policy set, no individual exchange has an incentive to deviate, all participants achieve positive payoffs, and aggregate welfare is maximised. By contrast, the model also identifies a stable equilibrium characterised by cost-minimising behaviour, which results in negative payoffs for all exchanges, illustrating a coordination failure driven by short-term incentives. These findings demonstrate that robust regulatory compliance can function as a coordination mechanism that aligns individual profit-seeking behaviour with long-run market stability and collective welfare.

The primary contribution of this thesis lies in providing a formal, strategic framework for evaluating crypto-asset regulation as an incentive structure rather than a purely exogenous constraint. By linking regulatory design to equilibrium outcomes, the analysis offers new insights into how coordinated compliance can enhance resilience, trust, and sustainability in cryptocurrency markets. The results have implications for policymakers designing regulatory regimes and for exchanges seeking strategies that balance profitability with long-term market credibility.

Chapter 1

Introduction

1.1 Motivation and context

Before the emergence of digital assets, investors primarily exchanged tangible commodities such as gold, silver, and oil goods valued for their physical properties and long-standing economic significance. Over time, financial markets evolved, giving rise to equity markets in which ownership was represented not through physical goods but through certificates and, eventually, entirely digital records. While early stock ownership was indeed documented on paper, modern securities have long since shifted to electronic registries; thus, stocks are better described as financial instruments rather than intangible assets in the same sense as cryptocurrencies. With the rapid global digitalization of the past decades, new categories of assets have emerged, among which cryptocurrencies represent one of the most transformative innovations.

When cryptocurrencies first appeared, they were widely dismissed as speculative experiments, or even outright scams, due to their novelty, technological complexity, and lack of regulatory oversight. Today, however, the landscape has changed dramatically. The cryptocurrency market now surpasses 1 trillion dollars in market capitalization, and major financial institutions, hedge funds, and even national governments hold various digital assets. Cryptocurrencies are increasingly seen as an innovative, high-growth investment class capable of yielding significant returns, which naturally attracts a growing appetite from both retail and institutional investors.

Cryptocurrency exchanges serve as the primary infrastructure enabling the buying and selling of these digital assets. Acting as intermediaries between market participants, they facilitate the flow of billions of dollars in capital each day. Yet the extent of their responsibility over user funds is still poorly understood by many, in part because regulatory frameworks governing exchanges are relatively new and remain unevenly implemented across jurisdictions. Unlike traditional financial institutions, cryptocurrency exchanges do not always follow unified operational standards, security practices, or compliance procedures. This inconsistency contributes to the perception of cryptocurrencies as risky or speculative and allows systemic vulnerabilities to persist.

External factors such as macroeconomic conditions, technological failures, regulatory decisions, and market sentiment intensify the volatility of digital assets. Many institutions continue to approach cryptocurrencies cautiously, citing the absence of mature regulatory structures that could mitigate risks arising from security breaches, operational mismanagement, or fraudulent activities. Exchanges compete fiercely to attract customers by listing a wide array of assets, but this openness also exposes them to threats such as money laun-

dering, market manipulation, and the listing of illegitimate or low-quality projects. Any major incident, whether an exchange collapse or a large-scale hack, diminishes investor confidence, reduces market participation, and negatively impacts the entire ecosystem due to the interconnected nature of cryptocurrency markets.

For these reasons, understanding how exchanges make decisions and how they respond to regulatory pressures is vital. A robust and well-designed global framework could significantly improve the stability, legitimacy, and long-term growth of the cryptocurrency market, encouraging broader institutional participation and safeguarding the interests of all investors.

1.2 Problem statement

Regulatory oversight of cryptocurrency markets remains in an early stage of development, and despite rapid industry growth, most jurisdictions still lack coherent and comprehensive frameworks capable of addressing the risks associated with digital asset trading. The European Union has introduced the MiCA framework with the intention of establishing a unified regulatory environment that applies consistently across all member states. MiCA sets out clear licensing requirements and operational standards that cryptocurrency exchanges must implement in order to access and operate within the EU market. While these regulatory measures aim to enhance consumer protection, strengthen market integrity, and reduce systemic risk, their real-world effectiveness has yet to be fully evaluated. It is crucial that such policies do not unintentionally hinder innovation or discourage new exchanges from entering the European market, as excessive regulatory burdens could limit competition and slow the development of the broader cryptocurrency ecosystem.

Within the framework of this thesis, our objective is to understand the impact of policies on cryptocurrency exchanges. In order to do that we construct a game theoretical model that captures the strategic behavior of multiple cryptocurrency exchanges when faced with the policy requirements outlined in MiCA. The model incorporates exchanges with varying incentives, competitive pressures, and internal strategies regarding compliance. By treating MiCA's regulatory requirements as optional elements within the model, we analyze the advantages and disadvantages of adopting each policy, as well as the consequences of non-adoption for both individual exchanges and the market as a whole. This approach allows us to examine how regulatory decisions influence exchanges' strategic choices, how compliance or non-compliance affects competitive dynamics, and to what extent the MiCA framework succeeds in promoting stability without suppressing innovation.

1.3 Structure of the Thesis

The remainder of this thesis is organized as follows. Chapter 2 provides background on cryptocurrency markets, exchange failures, cybersecurity incidents, and the regulatory landscape, with particular emphasis on MiCA and related global frameworks. Chapter 3 develops the conceptual and analytical framework, mapping regulatory policies to economic dimensions such as security, trust, market capitalization, and compliance costs. Chapter 4 presents the game-theoretical model, its assumptions, parameterization, and simulation design, and analyzes strategic outcomes including Nash equilibria and social optimality. Finally, Chapter 5 summarizes the main findings, discusses limitations of the model, and outlines directions for future research.

Chapter 2

Background

Cryptocurrencies represent a novel class of digital assets built upon decentralized ledger technologies, most notably blockchain systems. Unlike traditional financial assets, which rely on centralized authorities such as banks, governments, or regulatory bodies, cryptocurrencies operate through distributed networks of nodes that collectively verify and record transactions. The creation of Bitcoin in 2009 marked the first successful implementation of a decentralized digital currency. As new cryptocurrencies emerged, such as Ethereum with its ability to support programmable smart contracts, the cryptocurrency market evolved beyond digital money to encompass a broad array of applications including decentralized finance (DeFi), tokenized assets, and blockchain-based infrastructure services.

The cryptocurrency market is characterized by a high degree of price volatility. This volatility stems from its speculative nature, rapidly changing technological environment, and strong dependence on investor sentiment. Macroeconomic developments, regulatory announcements, global financial events, and technological breakthroughs can all influence market perception and cause significant fluctuations in asset prices. Although institutional participation from entities such as BlackRock, MicroStrategy, and various hedge funds has helped reduce some of the extreme volatility seen in the early stages of the market, particularly for large-cap assets such as Bitcoin and Ethereum, the effect is not uniform. Large institutions generally adopt long-term investment strategies, allocate capital in a structured manner, and follow risk-management procedures that differ greatly from the short-term, speculation-driven behavior typical of many retail traders. Their consistent accumulation of assets, increased liquidity provision, and reliance on regulated custodial services help stabilize price movements by reducing the impact of abrupt buy or sell pressures. Furthermore, institutional involvement signals a degree of legitimacy and confidence in the asset class, which can moderate panic-driven reactions during periods of uncertainty. Despite these stabilizing influences, cryptocurrencies remain susceptible to sudden and dramatic price movements. These fluctuations are often intensified by fragmented global regulation, inconsistent security practices among exchanges, and varying levels of market liquidity.

Market manipulation is another major concern within cryptocurrency. Practices such as pump-and-dump schemes and artificially inflated trading volumes can distort the fair value of assets, especially on exchanges that lack strict oversight or fail to implement proper compliance mechanisms. Furthermore, several historical failures of major cryptocurrency exchanges, whether due to managerial incompetence, insufficient security measures, or fraudulent activity, have resulted in severe financial losses and widespread panic among

investors. These incidents are directly relevant to this thesis, which investigates how inadequate operational safeguards many of which regulatory frameworks seek to mandate can lead to sharp declines in market prices, undermine investor confidence, and reinforce the perception of cryptocurrencies as unstable financial instruments.

The valuation of cryptocurrencies depends on several critical elements, including the technological potential of the underlying project, the level of trust within the community, the extent of network adoption, and the degree of speculative interest in the asset. For the most established cryptocurrencies, market capitalization is influenced not only by innovation but also by global economic conditions and shifts in collective sentiment. Although cryptocurrencies share certain characteristics with traditional investment assets such as commodities and equities, they face unique challenges resulting from their decentralized nature. Issues related to mainstream adoption, regulatory clarity, custodial security, and practical usability continue to influence market development and shape investor behavior.

Because the cryptocurrency industry is still young compared to traditional financial markets, regulatory guidance remains in a developmental phase. Policymakers around the world are actively working on frameworks, such as the European Union's MiCA regulation, intended to reduce systemic risks, strengthen investor protection, and minimize the likelihood of market failures caused by security breaches, operational mismanagement, or a lack of transparency. However, the fast-paced evolution of blockchain technology makes it difficult to establish consistent and harmonized regulation across jurisdictions. This ongoing process of regulatory maturation plays an essential role in understanding exchange behavior, identifying sources of market manipulation, and analyzing the strategic interactions that form the core of this thesis.

2.1 Related work

A key theoretical reference for the modelling of cybersecurity risk in this thesis is the framework developed by Gordon and Loeb [16], which demonstrates that security investment reduces the probability of a successful breach subject to diminishing marginal returns. Their model shows that initial security measures yield large reductions in expected loss, while additional investment becomes progressively less effective, implying that optimal security spending is bounded. The breach-probability function(4.2) used in this thesis adopts this core economic intuition by specifying security effectiveness as a decreasing, nonlinear function, but differs in its application and scope.

This intuition is further supported by Johnson et al. [19] , who model security investment decisions of cryptocurrency exchanges and show that optimal investment balances expected breach losses against operational costs, consistent with diminishing marginal returns to security spending. This work also helped shape the intuition underlying the specification and bounding of parameters in the functional forms.

2.2 Game theory basics

Game theory helps us study strategic interactions among rational decision makers whose outcomes depend on the choices they make as well as the choices of others. It has been applied in finance, economics, and regulatory analysis to examine coordination problems, incentives, and equilibrium behavior in environments characterized by interdependence and strategic uncertainty. In the context of cryptocurrency markets, game theory is par-

ticularly well suited to analyze how exchanges respond to regulatory constraints and competitive pressures.

2.2.1 Players

Players are the decision-making entities whose strategic choices jointly determine outcomes. Players are assumed to be rational in the sense that they seek to maximise their own objectives given the information available to them.

2.2.2 Strategies

A strategy specifies a complete plan of action for a player, describing how it will behave in every relevant situation of the game. Strategies can be deterministic or probabilistic combinations of actions.

2.2.3 Payoffs and utility function

The outcome of a game is evaluated through payoffs, which represent the utility or benefit each player derives from a particular combination of strategies. Payoffs typically incorporate profits, costs, risks, and other relevant economic considerations.

2.2.4 Social Optimum

The socially optimal outcome is defined as the strategy profile that maximises aggregate welfare across all players in the game. Social welfare is typically measured as the sum of individual payoffs or utilities, reflecting total economic surplus.

2.2.5 Nash Equilibrium

A central concept in game theory is the Nash equilibrium, which describes a strategy profile in which no player has an incentive to unilaterally deviate from its chosen strategy, given the strategies of all other players. At a Nash equilibrium, each player's strategy is a best response to the strategies of others.

Importantly, Nash equilibria does not need to be socially optimal 2.2.4. In many settings, individually rational behaviour can lead to outcomes that are collectively inefficient, giving rise to coordination failures or prisoners' dilemma–type outcomes.

2.2.6 Price of anarchy

The Price of Anarchy (PoA) measures the efficiency loss that arises when players act strategically and settle in the worst Nash equilibrium, compared to the social optimum. It captures how much collective welfare is sacrificed due to decentralised, self-interested behaviour.

$$PoA = \frac{SocialOptimum}{WorstNashEquilibrium} \tag{2.1}$$

A higher Price of Anarchy indicates a larger gap between individually rational behaviour and socially desirable outcomes. In economic and regulatory contexts, a high PoA signals the presence of coordination failures, externalities, or misaligned incentives that can lead to inefficient and unstable market outcomes.

2.2.7 Price of stability

While the Price of Anarchy focuses on the worst equilibrium, the Price of Stability (PoS) considers the best Nash equilibrium. It measures the smallest efficiency loss achievable among equilibrium outcomes.

$$PoA = \frac{SocialOptimum}{BestNashEquilibrium} \tag{2.2}$$

The Price of Stability provides insight into whether efficient outcomes are attainable without central coordination or enforcement. A Price of Stability close to one indicates that there exists at least one equilibrium that closely approximates the social optimum, even if other inefficient equilibria also exist.

2.3 Cybersecurity Incidents In Cryptocurrency Exchanges

Cybersecurity threats are a central concern that modern regulatory frameworks aim to address, as they represent one of the most significant risks facing cryptocurrency exchanges. These platforms store large volumes of digital assets and operate entirely in online environments, making them especially vulnerable to threats such as private-key theft, hot-wallet breaches, insider attacks, and smart-contract exploits. In the early stages of the cryptocurrency market, the absence of standardized regulatory requirements meant that security practices were largely dependent on the goodwill and technical maturity of individual exchanges. This lack of oversight contributed to several high-profile breaches that resulted in billions of dollars in losses, undermining investor trust and destabilizing market confidence. These incidents demonstrate the urgent need for regulatory frameworks that impose strong custody requirements, multi-layered security protocols, regular cybersecurity audits, and robust governance standards to reduce systemic vulnerabilities. The following major cases illustrate how insufficient security and limited regulatory oversight left exchanges exposed to catastrophic failures, and how modern policy measures could have significantly mitigated or prevented the resulting damage.

2.3.1 MT.Gox (2014)

The Mt. Gox breach remains one of the most influential and devastating cybersecurity failures in the history of cryptocurrency, resulting in the loss of approximately 850,000 BTC, valued at more than 450 million dollars at the time. The attack exploited a series of critical weaknesses in Mt. Gox's internal security practices, including its reliance on a single-signature hot wallet, the absence of effective transaction monitoring, and a lack of routine security audits. With no regulatory policies in place to mandate asset segregation, governance oversight, or operational controls, the exchange operated with significant vulnerabilities that allowed the theft to go undetected for an extended period. Under modern regulatory frameworks, requirements such as mandatory cold-storage policies, multi-layered custody protections, comprehensive operational risk assessments, and

formal incident-reporting obligations could have significantly mitigated or even prevented the scale of losses experienced by Mt. Gox.

2.3.2 Bitfinex (2016)

Bitfinex experienced a major security breach that resulted in the loss of approximately 72 million dollars worth of Bitcoin. The incident was caused by vulnerabilities within the exchange's multi-signature wallet architecture, which required coordination with a third-party custodian. At the time, regulatory authorities did not enforce standardized security protocols or clearly define responsibilities within custodial arrangements, leaving Bitfinex with significant gaps in access controls and operational oversight.

2.3.3 KuCoin (2020)

KuCoin suffered a significant security breach in which attackers stole more than 280 million dollars in various cryptocurrencies after obtaining the private keys that controlled the exchange's hot wallets. The incident exposed serious weaknesses in KuCoin's key-management practices and insufficient protection of critical infrastructure components. Regulatory requirements mandating the use of hardware security modules (HSMs), multi-layered key-management protocols, regular penetration testing, and comprehensive cybersecurity audits could have substantially strengthened the exchange's defenses and greatly reduced the likelihood of private-key compromise.

2.3.4 ByBit (2025)

In early 2025, Bybit experienced one of the largest cryptocurrency exchange breaches ever recorded, with an estimated 1.5 billion dollars stolen from one of its custodial wallets. The incident did not stem solely from internal mismanagement but was enabled by a sophisticated exploitation of a third-party wallet provider interface (SafeWallet). Attackers manipulated the interface to make malicious transactions appear legitimate, allowing unauthorized transfers to be signed and executed without triggering immediate security alerts. Although Bybit was a major global exchange with significant technological resources, the breach exposed critical vulnerabilities in its custody infrastructure, particularly in third-party dependency management. The incident underscored the necessity for exchanges to conduct thorough due diligence on external service providers, maintain strict segregation of critical system components, perform routine security audits, and ensure that custody arrangements adhere to robust risk-management standards designed to prevent such large-scale compromises.

2.4 Operational Risk and Resilience In Cryptocurrency Exchanges

Operational risk is one of the most pervasive and damaging threats facing cryptocurrency exchanges, arising from failures in governance, internal controls, fund management, communication practices, and overall organizational oversight. In contrast to cybersecurity incidents, which are typically caused by external attacks, operational failures originate within the exchange itself and are often the result of mismanagement, inadequate internal policies, or insufficient regulatory supervision. Because exchanges function as custodians

of significant volumes of customer assets, weaknesses in operational resilience can lead to catastrophic financial losses and long-term erosion of investor trust. The following major cases demonstrate how poor governance, financial mismanagement, and the absence of proper internal controls exposed exchanges to severe losses.

2.4.1 FTX Collapse (2022)

The collapse of FTX stands as one of the most significant operational failures in the history of the cryptocurrency sector, with an estimated 8 billion dollars in customer funds unaccounted for. The downfall was driven by internal mismanagement and the absence of basic governance structures. Investigations revealed that FTX commingled customer assets with those of its affiliated trading firm, Alameda Research, allowing Alameda to use client funds for proprietary trading and high-risk investments. When market conditions deteriorated and Alameda's positions collapsed, the exchange no longer had sufficient liquid reserves to meet customer withdrawals. FTX also lacked adequate risk-management systems, proper accounting practices, and any meaningful level of financial oversight. While capital adequacy was not a formally regulated requirement at the time, FTX's inability to maintain sufficient liquidity or segregate client assets effectively meant that it could not compensate customers once losses accumulated.

2.4.2 QuadrigaCX (2019)

The QuadrigaCX incident resulted in approximately 190 million dollars in inaccessible customer assets after the exchange's founder, who held sole control over the private keys and critical operational systems, unexpectedly died in 2018. This unprecedented situation revealed severe governance failures, including the absence of internal oversight, the lack of multi-party authorization for asset access, and a complete failure to segregate or properly secure customer funds. The collapse of QuadrigaCX thus stands as a clear example of how inadequate oversight and concentrated control can lead to catastrophic operational failure in cryptocurrency exchanges.

2.5 Regulatory Frameworks for Cryptocurrency Exchanges

Regulatory frameworks for cryptocurrency exchanges play a critical role in shaping the structure, stability, and legitimacy of the digital-asset market. These frameworks outline how an exchange must operate within a specific jurisdiction and define the standards it must meet when offering services or handling customer funds. Compliance with these requirements is essential for obtaining a license to operate in a region, and such licensing brings obligations related to transparency, disclosure, safeguarding of customer assets, and adherence to consumer-protection, anti-money-laundering (AML), and counter-terrorist-financing (CFT) rules. The central objective of these frameworks is to ensure that service providers maintain clear governance structures, implement robust security measures, and operate in a manner that builds investor confidence and public trust in the cryptocurrency ecosystem.

At the global level, the Financial Stability Board (FSB) has introduced a high-level regulatory framework for crypto-asset activities and stablecoin arrangements, offering recommendations on licensing, risk management, governance, custody, transparency, operational resilience, and cross-border cooperation. Additional regulatory initiatives, such as know-

your-customer (KYC) and enhanced due-diligence requirements, aim to reduce anonymity, monitor crypto-asset flows, and mitigate risks associated with illicit finance. Collectively, these frameworks strongly influence market behavior by requiring exchanges to adopt protocols similar to those used in traditional financial institutions.

In jurisdictions lacking such regulations, markets tend to be fragmented, oversight is weak, and users may rely on unregulated intermediaries, which significantly increases the likelihood of fraud, mismanagement, and consumer harm. Conversely, when exchanges operate under consistent regulatory expectations, they are more likely to invest in strong security infrastructure, establish effective internal controls, and provide transparent services. These practices reduce the frequency and severity of operational failures and help prevent insolvency or fraudulent collapses. Regulation therefore functions both as a deterrent against misconduct and as an incentive for responsible market behavior.

However, regulatory approaches differ widely across countries. Some jurisdictions classify crypto-assets as commodities, others as securities, and a few have imposed outright bans. This global fragmentation complicates cross-border operations and enables regulatory arbitrage, where exchanges relocate to jurisdictions with weaker compliance standards, thereby increasing systemic risk [21].

2.5.1 MiCA

The MiCA framework is the European Union's collective regulatory response aimed at creating a unified and consistent system of oversight for cryptocurrency exchanges across all EU member states. Its main goals are to reduce the risks of money laundering and illicit financial activity, increase traceability within digital-asset markets, and improve the operational standards of exchanges by requiring them to implement the policies set out in the regulation. MiCA also addresses the long-standing issue of regulatory fragmentation in Europe by replacing disparate national rules with a single, harmonized framework that all exchanges must follow, significantly reducing bureaucratic barriers and simplifying cross-border operations. Entering into force between June and December 2024, the framework has quickly become a foundational regulatory model designed to bring crypto-assets into a transparent, secure, and well-supervised financial environment, positioning the EU as a global leader in structured digital-asset regulation. Beyond financial-crime prevention and operational oversight, MiCA's core purpose is to provide legal clarity for both crypto-asset issuers and service providers.

2.5.2 Global Frameworks

Global regulatory frameworks for cryptocurrency vary widely across jurisdictions, resulting in a fragmented landscape that strongly influences how exchanges operate internationally. In the United States, for example, regulatory authority is distributed across several agencies, including the Securities and Exchange Commission (SEC), which oversees securities of the markets, the Commodity Futures Trading Commission (CFTC), which regulates commodity derivatives and certain digital asset activities, and the Financial Crimes Enforcement Network (FinCEN), which enforces AML and CFT obligations. These agencies interpret digital assets under different legal classifications, leading to overlapping, inconsistent, and sometimes conflicting regulatory requirements. In contrast, countries such as Japan and Singapore have implemented dedicated licensing frameworks for cryptocurrency exchanges that resemble the goals of the MiCA framework by emphasizing consumer protection, operational resilience, and strong AML and KYC controls. Meanwhile, other

jurisdictions including several offshore financial centers maintain more permissive regulatory environments that attract exchanges seeking lighter compliance burdens, contributing to regulatory arbitrage and uneven global standards. International bodies such as the FSB have attempted to increase global coordination by issuing high-level guidelines on AML and KYC obligations, stablecoin oversight, market integrity, and tax transparency.

However, the adoption and enforcement of these standards vary significantly across regions, and no unified global regulatory framework currently exists. This divergence creates structural vulnerabilities, as weaknesses in one jurisdiction can spill over into global markets through interconnected exchanges and capital flows. Nevertheless, the growing involvement of major international institutions indicates a gradual movement toward more harmonized, risk-sensitive, and transparent global regulation, an evolution that is likely to strengthen market stability, improve investor protection, and enhance the overall credibility of the cryptocurrency industry.

2.5.3 User Behavior and Market Confidence

The cryptocurrency market is still relatively young and remains highly sensitive to incoming information, whether positive or negative. News events can trigger rapid behavioral responses among investors such as panic selling during crises or fear of missing out (FOMO) during bullish developments which in turn influence overall market capitalization. Historical events illustrate this dynamic clearly: the Mt. Gox collapse in 2014 caused Bitcoin to fall by more than 30%, while the FTX bankruptcy in 2022 caused the total cryptocurrency market capitalization to fall by more than 20% within a day as market confidence deteriorated (see Figure 2.1).

These cases show that negative news involving exchange failures, hacks, or operational mismanagement not only erodes trust in the affected platform but also creates contagion effects across the broader crypto ecosystem, as retail users fear widespread instability and rush to liquidate their holdings. On the other hand, positive developments such as enhancements in exchange security, the introduction of strong regulatory frameworks, improvements in custodial infrastructure, or institutional adoption tend to generate bullish sentiment and increase market participation, thereby elevating overall market capitalization. For example, Bitcoin's sustained upward movement during 2020–2021 coincided with the emergence of institutional-grade custody solutions and the entry of major financial institutions, illustrating how greater operational maturity strengthens investor confidence. User trust in exchanges is closely tied to perceptions of platform security, transparency, and usability [27]; industry surveys consistently show that investors prioritize robust security features and an intuitive user experience when selecting where to buy, trade, or store digital assets. Exchanges that maintain high operational resilience, publish audits, or comply with established regulatory standards generally generate greater revenue because users feel safer entrusting their assets to well-governed platforms. Conversely, exchanges with poor management or repeated incidents of failure quickly lose their user base, demonstrating the direct relationship between trust, market participation, and market stability.

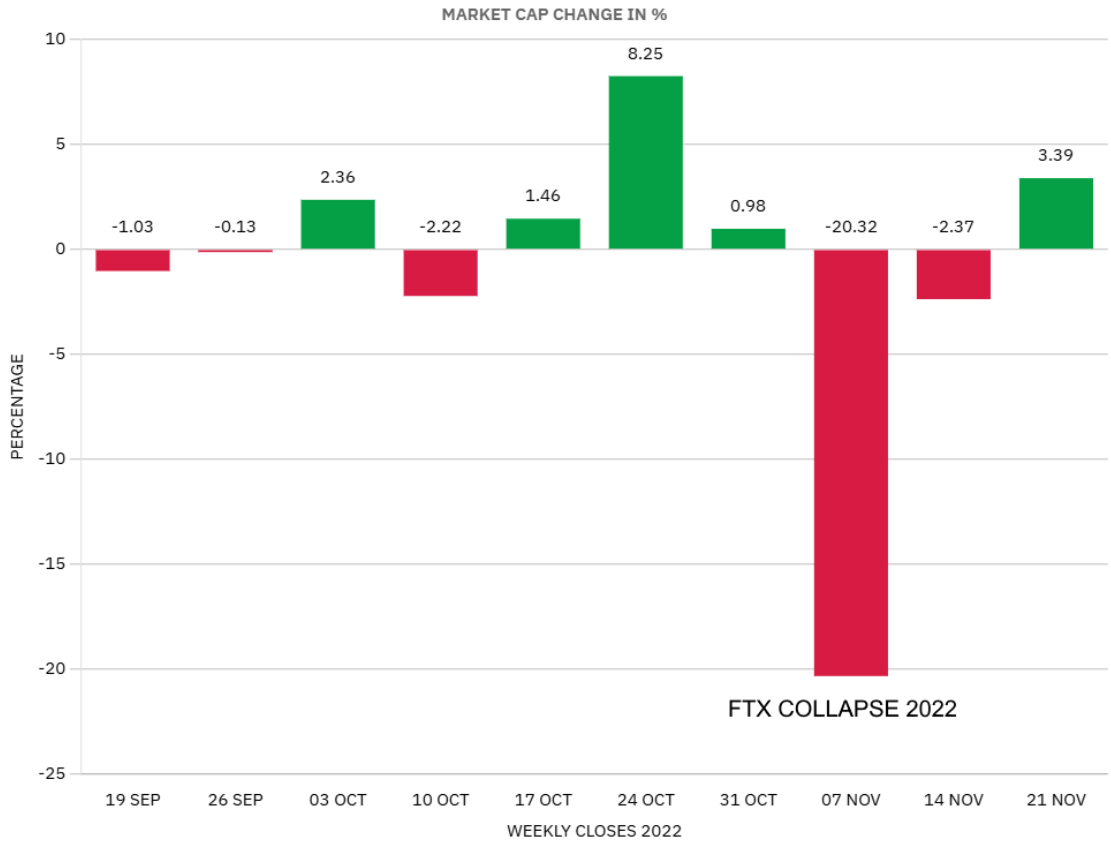

Figure 2.1: Evolution of market capitalization around the FTX collapse

Chapter 3

Conceptual Framework

In this thesis, a focused set of MiCA regulatory policies is examined to analyze how European regulatory conditions affect cryptocurrency exchanges and market dynamics, focusing on specific components of the regulation that exert distinct and measurable effects on cryptocurrency exchanges. These policies are examined collectively to evaluate their influence on exchange behavior, investor trust, financial security, and overall market performance. Although MiCA is a comprehensive regulatory framework, this model concentrates on the policies most relevant to how exchanges strategically operate within a regulated environment. Each selected policy is assigned a quantitative score that represents its impact across four key dimensions: financial security, user trust, market-cap influence, and compliance cost. These quantified mappings allow the regulatory requirements to be integrated into a game-theoretical framework in which multiple exchanges act as strategic players, choosing whether to implement or avoid each policy. The resulting payoffs reflect how these policy decisions influence an exchange's profitability, competitiveness, and ability to attract and retain users. By translating regulatory obligations into measurable strategic variables, the model assesses whether rational, profit-maximizing exchanges would independently adopt these policies and identifies the conditions under which regulation functions as an incentive, deterrent, or coordinating mechanism. This approach provides a systematic way to evaluate how MiCA's policy architecture may shape real-world exchange behavior and contribute to the stability, transparency, and credibility of the broader cryptocurrency market.

3.1 Mapping Policies to Security, Trust and Market Growth

3.1.1 Licensing Requirements

The Licensing Requirement of MiCA establishes the full legal framework for the authorization of cryptocurrency exchanges and other crypto-asset service providers (CASPs) operating within the European Union (EU). This policy mandates that no firm may legally offer crypto-asset services in the EU without first obtaining regulatory authorization. To obtain a license, firms must submit a comprehensive application outlining their business model, the specific crypto-asset services they intend to provide, and how their operations will be organized and controlled. Regulators assess whether the applicant demonstrates sound governance arrangements, including clearly defined management responsibilities, effective internal controls, and robust compliance frameworks, particularly with respect to AML and consumer protection requirements. In addition, firms must prove that their senior

management satisfies the fit and proper standard, meaning that directors and executives possess the necessary experience, integrity, and professional competence. These safeguards are designed to ensure that authorized firms do not pose undue risks to consumers or to the wider financial system. Once authorization is granted by a competent authority in one EU Member State, passporting rights apply, allowing the exchange to operate legally across all EU jurisdictions without the need for additional national licenses.

Market Cap

With respect to market capitalization, the licensing requirement influences crypto markets mainly by shaping market access and reinforcing institutional trust, rather than through direct effects on prices. Recent research indicates that mandatory authorization reduces the number of operating exchanges, particularly smaller firms, due to higher compliance, governance, and capital obligations. This contraction can produce short-term declines in regional trading activity and modest downward pressure on regional market capitalization, while leaving global market capitalization largely unchanged. Over longer terms, licensing is associated with measurable improvements in market quality. Weakly regulated cryptocurrency exchanges are more exposed to fraud, manipulation, liquidity deterioration, and price instability, whereas stronger regulatory frameworks support market integrity and more stable liquidity conditions [15]. By raising entry thresholds and redirecting activity toward well-capitalized and compliant platforms, licensing reduces fragmentation and liquidity instability while strengthening supervisory visibility and systemic-risk oversight. Although speculative retail participation may decline initially, regulated market environments are substantially more attractive to institutional investors, who are typically restricted from engaging with unlicensed venues. As EU-authorized platforms gain regulatory legitimacy, capital inflows from banks, funds, and payment institutions are more likely to be stable and persistent, supporting market-capitalization stabilization rather than sustained contraction, particularly for major crypto-assets such as Bitcoin and Ethereum. Nevertheless, while licensing strengthens liquidity conditions, transparency, and risk mitigation, it does not generate a direct or quantifiable causal pathway to long-term growth in total global crypto market capitalization. Its effect is therefore best characterized as structurally important but indirect. On this basis, the policy is assigned an influence score of 7/10, reflecting a moderate-to-high impact on market capitalization operating through institutional and structural channels rather than direct valuation effects.

Customer Trust

In terms of user confidence, the licensing requirement under MiCA has a strongly positive effect on how both retail and institutional users assess the safety and reliability of crypto exchanges. Regulatory authorization signals that an exchange has met enforceable minimum standards relating to governance, thereby reducing perceived risks of fraud, misconduct, and sudden platform failure. Evidence suggests that sound regulatory frameworks and oversight are associated with expanded participation in formal financial systems and greater financial inclusion [23] [4]. Similar dynamics apply in crypto markets. When exchanges are supervised, and subject to sanctions or license withdrawal, users are more willing to place larger balances on the platform and to retain assets for longer periods. Licensing also strengthens trust by clarifying accountability. In lightly regulated or offshore environments, users often face limited recourse when problems arise, which contributes to persistent trust deficits. Under MiCA, authorized exchanges must comply with conduct-of-business, safeguarding, and complaints-handling requirements, and regu-

latory bodies are empowered to intervene when harmful practices emerge. This aligns the client-exchange relationship more closely with regulated financial intermediation, which is an important condition for retail adoption of digital financial services. For institutional participants, authorization is frequently a binding prerequisite imposed by internal risk policies, meaning that many banks, funds, and payment institutions cannot interact with unlicensed exchanges. The existence of an EU-wide licensing framework therefore enables these institutions to treat authorized exchanges as legitimate and trustworthy counterparties, reinforcing perceptions of safety among retail users. At the same time, licensing does not eliminate all sources of risk. However, regulatory authorization does not eliminate all sources of risk: even licensed exchanges can mislead customers about the extent of regulation and face operational or product-related risks, potentially creating unrealistic expectations of protection [26]. A subset of highly risk-tolerant users may also prefer unregulated venues offering greater flexibility or prohibited features. Nevertheless, these limitations only partially offset the overall effect. In a market environment characterized by historically low baseline trust and elevated fraud and insolvency risks, MiCA licensing represents one of the most effective mechanisms for strengthening confidence in crypto exchanges. On this basis, the licensing requirement is assigned a signed customer-trust score of 8/10, reflecting a very strong, though not absolute, positive impact on users' willingness to entrust assets to licensed exchanges.

Security

When assessed through the scope of cybersecurity outcomes, the policy does not exert a direct or independent effect on the technical security of crypto exchanges. Licensing focuses primarily on governance arrangements, which supports institutional soundness but does not prescribe specific information and communication technology (ICT) safeguards such as encryption standards, network architecture, vulnerability management, or incident-response capabilities. Authorization and supervisory status alone do not materially reduce the likelihood of cyber incidents, since even highly regulated financial institutions and public authorities continue to experience significant breaches. Licensing, in isolation, does not meaningfully alter an exchange's attack surface, vulnerability profile, or defensive capacity in the absence of prescriptive ICT-security rules. On this basis, the licensing requirement is assigned a signed security score of 0/10, reflecting the absence of a demonstrable, independent impact on cybersecurity outcomes. A non-zero score would only be warranted if licensing itself imposed explicit technical security obligations.

Compliance Costs

From an operational standpoint, it imposes a substantial compliance burden on crypto exchanges. The authorization process entails significant upfront expenditure related to legal and regulatory fees, governance restructuring, compliance staffing, internal control frameworks, and ongoing supervisory reporting. Unlike informal or fragmented registration approaches that previously existed in parts of the EU, MiCA authorization requires exchanges to submit extensive documentation covering business models, operational structures, IT systems, AML arrangements, safeguarding mechanisms, and governance frameworks. PwC (2025) [25] notes that the implementation of the MiCA has established a harmonized yet comprehensive regulatory framework for CASPs, requiring firms to make significant organizational and operational changes to comply with new authorization, governance, and supervisory requirements, potentially increasing compliance costs and the need for ongoing advisory and oversight engagement. For exchanges transitioning from

lightly regulated or offshore models, costs tend to cluster toward the upper end of this range due to the need for comprehensive governance and compliance rebuilds. In addition to initial authorization, licensing generates material recurring costs. Exchanges must maintain permanent compliance functions, risk management and internal audit capabilities, regulatory reporting systems, and continuous engagement with supervisors. These recurring obligations constitute a fixed structural cost of operating in the EU market and represent a particularly high barrier to entry for smaller or early-stage exchanges. Moreover, once authorized, EU passporting mitigates regulatory duplication as exchanges expand across Member States, preventing compliance costs from scaling linearly with geographic reach. On balance, the licensing framework represents a high but economically survivable compliance obligation. Accordingly, it is assigned a compliance cost score of 7/10, corresponding to typical first-year costs in the low-to-mid single-digit millions of euros and recurring annual costs that materially affect profitability without being structurally prohibitive.

3.1.2 Operational Resilience and Cybersecurity (DORA)

DORA is a stand-alone EU regulation and is not formally part of MiCA; however, it applies directly to cryptocurrency exchanges operating within the EU financial regulatory framework. Exchanges authorized under MiCA are required to comply with DORA's digital operational resilience and cybersecurity obligations from 2025 onward. While MiCA governs market conduct, licensing, and the provision of crypto-asset services, DORA focuses specifically on the resilience and security of the ICT systems that support those activities.

The DORA framework establishes harmonized security practice standards across the EU, requiring firms to implement robust ICT risk management frameworks with clear governance, defined roles and responsibilities, and management-level accountability. It mandates preventive and detective security controls, including access management, encryption, monitoring, logging, backup and recovery procedures, as well as secure software development and change management. In addition, DORA introduces compulsory incident classification and reporting, alongside regular operational resilience testing such as penetration testing and scenario-based exercises. Together, these measures ensure that cryptocurrency exchanges can effectively prevent, detect, respond to, and recover from severe ICT-related disruptions, making digital operational resilience an integral and mandatory element of the EU crypto-regulatory architecture.

Market Cap

DORA affects crypto-market capitalization by reducing extreme operational risk and strengthening institutional confidence, rather than by influencing prices, trading volumes, or token supply directly. Cyber and ICT-related disruptions are increasingly treated as sources of systemic financial risk, as major incidents can halt exchange operations, disrupt liquidity provision, widen spreads, and rapidly erode investor confidence. Event-study evidence from traditional financial markets consistently shows that significant cyber incidents generate negative abnormal returns, persistent liquidity deterioration, and heightened volatility, demonstrating that cyber risk is directly reflected in asset valuations [20] [14]. By imposing minimum standards for ICT risk management, resilience testing, incident reporting, and third-party oversight, DORA seeks to reduce both the likelihood and severity of such tail events in EU-based crypto markets. For exchanges and custodial infrastructures, this influences market-capitalization dynamics through two principal mechanisms.

First, improved operational resilience lowers downside risk by reducing the probability of catastrophic hacks or prolonged outages that can trigger confidence-driven sell-offs. Second, enforceable resilience standards increase the acceptability of EU-regulated trading venues for banks, asset managers, and payment institutions whose participation is highly sensitive to cyber and operational risk. At the same time, DORA does not regulate token issuance, determine valuation mechanisms, or directly stimulate investor demand. Its contribution to market capitalization therefore operates through risk-premium compression and infrastructure reliability rather than through expansionary market forces. On this basis, Operational Resilience and Cybersecurity under DORA is assigned an influence score of 4/10, reflecting a meaningful but clearly indirect effect that is primarily protective in nature rather than growth-generating.

Customer Trust

Customer trust is materially strengthened by DORA because it directly targets operational and cybersecurity risks that have historically undermined confidence in the sector. Large-scale hacks, system outages, and prolonged service disruptions have fostered a perception that exchanges are technologically fragile and that access to client assets can be lost without warning. DORA enhances confidence in the stability of exchanges by reducing the risk of systemic technology failures and prolonged service disruptions. This shift in risk perception makes customers more willing to keep assets on exchanges rather than relying exclusively on self-custody or dispersing balances across multiple platforms for defensive reasons. Trust is further reinforced by the way DORA reshapes expectations around crisis management. Mandatory incident reporting, recovery planning, and resilience testing signal that operational disruptions must be handled within predefined regulatory processes, reducing fears of sudden, opaque service failures and prolonged loss of account access. These assurances are particularly salient for institutional and professional clients, whose participation is highly sensitive to downtime and operational continuity. Many such institutions require counterparties to demonstrate audited cybersecurity and business-continuity arrangements, meaning that DORA directly expands the pool of exchanges that can be regarded as institutionally credible custodians. At the same time, the trust effects of DORA remain bounded. Enhanced cybersecurity standards do not shield users from market risk, governance failures, or legal uncertainty in insolvency scenarios, and most customers cannot directly observe the quality of technical controls in place. Perceptions of safety therefore continue to depend on regulatory credibility, enforcement, and past performance. In light of these considerations, Operational Resilience and Cybersecurity under DORA is assigned a signed customer-trust score of 7/10, reflecting a strong but not dominant positive influence on users' willingness to entrust assets to regulated exchanges.

Security

DORA materially strengthens the technical security of cryptocurrency exchanges by directly targeting ICT vulnerabilities. Unlike governance-oriented rules, DORA imposes requirements on how exchanges must manage the full lifecycle of ICT risk across systems that underpin wallet infrastructure, custody platforms, trading engines, and customer-facing APIs. For crypto exchanges, this translates into formalized controls over access to private-key environments, segregation of critical systems, continuous monitoring of custody and trading infrastructure, and structured incident-response and recovery processes designed to limit asset loss and service disruption during cyber incidents. Mandatory

resilience testing, including threat-led penetration testing, is particularly relevant in a crypto context where attackers frequently target hot-wallet configurations and signing processes. By mandating the documentation, testing, and auditability of ICT risk controls, DORA strengthens oversight of operational vulnerabilities, including those related to system access, interfaces, and third-party dependencies. International policy assessments consistently emphasize that enforceable cybersecurity standards and supervisory oversight are essential to mitigating system-wide cyber risk, particularly in highly interconnected financial systems that rely on shared infrastructures and service providers [18] [11]. By embedding these principles into binding EU law, DORA substantially lowers both the probability and expected severity of successful cyberattacks affecting crypto exchanges, not by eliminating risk entirely but by raising baseline cyber hygiene and improving detection, containment, and recovery when incidents occur. Outcomes nevertheless remain dependent on implementation quality, legacy architecture, and the evolving threat landscape, and supervisors themselves acknowledge the need for ongoing adaptation as attack vectors evolve. For these reasons, DORA is assigned a security score of 9/10, reflecting an exceptionally strong but not absolute improvement in exchanges' resilience to cyber breaches.

Compliance Costs

Compliance with DORA is among the most financially demanding obligations faced by EU-based cryptocurrency exchanges because it targets the technology stack that underpins exchange operations rather than legal form or organizational structure. DORA requires sustained investment in the ICT infrastructure used for custody, trading, risk monitoring, and client access. Compliance entails upgrading wallet and key-management environments, implementing continuous monitoring across trading engines and APIs, deploying advanced detection and response tooling, conducting recurring penetration and resilience testing, and establishing robust backup, recovery, and incident-management capabilities. These obligations extend to the management of critical third-party providers, including cloud infrastructure, custody-technology vendors, and managed security services, which are central to most exchange operating models. Industry analyses suggest that enhanced operational-resilience and cybersecurity requirements can entail substantial compliance and investment costs for firms. Media and consultancy assessments indicate that regulatory-driven cybersecurity and resilience spending may amount to several thousand euros per employee annually in some organizations, with aggregate global expenditures on cybersecurity and operational resilience reaching into the hundreds of billions [13]. Survey evidence cited by Deloitte indicates that most financial institutions expect initial DORA implementation budgets in the €2–5 million range, even outside the systemic tier [8]. For a EU-focused crypto exchange employing several hundred staff, realistic implementation costs therefore lie in this range, followed by recurring annual expenditures ranging from high six figures to several million euros for specialist staff, tooling, testing, audits, and regulatory reporting. Larger or multi-jurisdictional exchanges can see these costs scale rapidly, reflecting their more complex architectures and heavier reliance on third-party ICT providers. In addition, DORA's enforcement framework, including administrative fines of up to 2% of global annual turnover, creates strong incentives for conservative over-investment in compliance, further increasing effective costs. Taken together, these factors justify assigning DORA a compliance cost score of 8/10.

3.1.3 Outsourcing Controls

The outsourcing controls under MiCA establish a dedicated regulatory framework governing how exchanges may delegate critical or important operational functions to third-party service providers. The objective of this framework is to ensure that outsourcing arrangements do not weaken an exchange's operational resilience and supervisory oversight. Under these rules, firms remain fully responsible and legally accountable for all outsourced activities, even where services are performed by external providers. Outsourcing must be governed by formal written agreements that clearly define the rights and obligations of both parties, including regulatory and audit access rights, data protection safeguards, business continuity measures, and termination provisions. The framework further requires that outsourcing arrangements do not impair the quality of internal controls, risk management, or compliance with MiCA obligations, and explicitly prohibits structures that would reduce an exchange's operational effectiveness. Where critical functions are outsourced, notification to the competent authority is required, together with evidence that risks relating to concentration, dependency, and third-party failure are adequately managed. These requirements are particularly significant for crypto markets, where reliance on cloud infrastructure, custody technology providers, data analytics services, market-making systems, and cybersecurity vendors is widespread. As a result, MiCA's outsourcing framework operates as a direct regulatory constraint on how technological and operational infrastructures within the crypto industry are structured.

Market Cap

Outsourcing controls influence crypto-market capitalization primarily by shaping operational stability, managing concentration risk, and supporting confidence in exchange infrastructure, rather than by affecting prices, trading incentives, or token supply directly. Regulators increasingly identify third-party and outsourcing dependencies as sources of systemic operational risk, particularly where large segments of the financial system rely on a small number of cloud providers, custody-technology vendors, or core infrastructure services. This risk is especially pronounced in crypto markets, where the same providers often support wallet infrastructure, trading engines, and custody systems for dozens of exchanges simultaneously. A failure at a single critical provider can therefore disrupt trading and asset access across multiple platforms at once. By requiring exchanges to govern outsourcing relationships through detailed contractual safeguards, audit rights, and supervisory visibility, the outsourcing framework seeks to limit excessive concentration and reduce the likelihood of correlated operational failures. Preventing such cascading disruptions helps protect market-wide liquidity and reduces the probability of abrupt confidence shocks that could trigger sharp, system-wide declines in market capitalization. Evidence from financial markets shows that exposure to operational and cyber risks, including third-party dependencies, is associated with negative valuation effects, indicating that operational risk is priced by investors [14]. At the same time, outsourcing controls also shape market structure. The need to maintain compliant contractual frameworks and oversight mechanisms raises fixed operating costs, which smaller exchanges, particularly those heavily dependent on external infrastructure, may struggle to absorb. This can encourage consolidation, shifting activity toward larger, better-capitalized platforms. While this may temporarily reduce competitive intensity or regional liquidity, it tends to improve average platform reliability and resilience, supporting confidence over the medium term. Because these effects operate through operational risk reduction and infrastructure stability rather than through direct demand or supply channels, the impact on total global

crypto market capitalization is meaningful but inherently constrained. On this basis, outsourcing controls are assigned a market-capitalization influence score of 4/10, reflecting a moderate, protective effect that stabilizes existing market value rather than actively driving new capital formation.

Customer Trust

Outsourcing controls contribute to customer trust in crypto exchanges by reducing uncertainty around hidden third-party dependencies and clarifying responsibility when operational failures occur. In the absence of clear rules, users often have little insight into whether exchanges rely on external providers for custody infrastructure, trading engines, data storage, or cybersecurity, and uncertainty over these arrangements can undermine confidence when disruptions arise. By requiring exchanges to retain full legal accountability for outsourced critical functions and to ensure audit access, data protection, and business-continuity safeguards, the outsourcing framework strengthens the perception that there is always a clearly identifiable entity responsible for service reliability. This reduces concerns that losses or access disruptions will be attributed to remote or lightly regulated vendors, increasing users' willingness to keep assets on a single platform rather than dispersing them purely for defensive reasons. Trust is further supported through the management of concentration risk. Even without technical expertise, retail users react strongly to widely reported cloud outages or vendor failures that affect multiple exchanges simultaneously. By obliging exchanges to monitor and mitigate excessive dependence on individual providers, outsourcing controls lower the perceived likelihood that a single external failure will simultaneously restrict access to funds across several platforms. This contributes to a more stable trust environment, particularly during periods of market stress when operational continuity becomes a primary concern. At the same time, the trust effects of outsourcing controls remain limited in visibility. Most users do not directly observe contractual safeguards, audit rights, or concentration metrics, and therefore infer their value indirectly through regulatory credibility rather than immediate, tangible protections. Moreover, these controls do not eliminate all technological risk, nor do they address fraud, insolvency, or market losses. For these reasons, outsourcing controls are assigned a signed customer-trust score of 5/10, reflecting a moderate but clearly positive effect on users' willingness to entrust assets to crypto exchanges, materially supportive of institutional credibility yet less salient to customers than more direct safeguards such as licensing or asset segregation.

Security

Outsourcing controls have a clear positive effect on exchange resilience against supply-chain compromise. European threat landscape assessments highlight that attackers rapidly exploit vulnerabilities including those in software dependencies and third-party service providers that serve multiple firms amplifying risk across interconnected digital infrastructures [12].In crypto markets, where exchanges rely heavily on external infrastructure, these dependencies create concentrated points of failure. By requiring exchanges to identify which outsourced activities are critical and to retain full responsibility for those functions, MiCA's outsourcing framework strengthens due diligence and governance around third-party services. Mandatory risk assessments, audit and access rights, data-protection provisions, and business-continuity obligations reduce the likelihood that severe vendor-level weaknesses persist unnoticed or unmanaged. Guidance from the European Banking Authority (EBA) emphasizes that firms must integrate cybersecurity considerations into

vendor selection, contractual design, ongoing monitoring, and exit strategies when outsourcing critical functions, and must maintain up-to-date visibility over where and how those functions are performed [10]. While outsourcing controls cannot prevent attackers from targeting large providers, they materially raise the baseline of monitoring and enforceability around third-party security. On this basis, outsourcing controls are assigned a signed security score of 6/10.

Compliance Costs

MiCA's outsourcing controls impose a moderate but clearly material financial burden on crypto exchanges. The main cost drivers arise from the need to establish or enhance third-party risk management (TPRM) frameworks, maintain detailed outsourcing registers, and review, renegotiate, or remediate contracts with key service providers. Because the outsourcing framework is closely aligned with EBA outsourcing guidelines, exchanges are effectively required to adopt governance, documentation, and oversight standards comparable to those applied to banks and investment firms. In practice, compliance requires mapping all outsourced activities, identifying which functions qualify as critical, maintaining a comprehensive and regularly updated outsourcing register, embedding mandatory contractual provisions such as audit rights, business-continuity safeguards, and exit clauses, and conducting ongoing monitoring of third-party risks. Although these requirements are significantly less costly than DORA's technology-intensive obligations, they are far from negligible. MiCA-driven outsourcing compliance typically involves a one-off implementation and contract-remediation phase in the estimate of high hundreds of thousands to low millions of euros, followed by recurring annual costs in the low hundreds of thousands of euros for dedicated staff, TPRM tooling, vendor assessments, and ongoing oversight. While MiCA-specific cost studies remain limited, industry analyses indicate that outsourcing governance and third-party risk oversight represent recurring and resource-intensive compliance functions across financial services, rather than marginal administrative tasks, as reflected in the scale and growth of the global TPRM market [7]. On this basis, outsourcing controls are assigned a compliance cost score of 5/10, reflecting a mid-range but non-trivial burden.

3.1.4 Capital and Own-Funds Requirements

The capital and own-funds requirements under MiCA establish the prudential financial standards that exchanges must meet to obtain and maintain authorization in the EU. These rules require exchanges to hold a minimum level of own funds on a continuous basis, calibrated to the nature and scope of the services they provide. The objective of this framework is to ensure financial resilience, enabling firms to absorb operational and market losses and to meet their obligations to clients even under stress conditions. At both the authorization stage and on an ongoing basis, entities must demonstrate that their capital is sufficient in quantity and quality, typically consisting of paid-up share capital, retained earnings, or other high-quality capital instruments. Where capital levels fall below the required thresholds, firms are required to take immediate remedial action, and supervisors may impose restrictions, require recapitalization, or withdraw authorization. Substantively, MiCA's own-funds framework imports a prudential logic comparable to that applied in traditional banking and investment-firm regulation, while adapting it to the operational, custody, and technological risks specific to crypto markets. In doing so, it marks a shift away from the historically lightly capitalized structure of many crypto

exchanges toward a model of regulated financial intermediation supported by explicit loss-absorbing buffers.

Market Cap

Capital and own-funds requirements influence crypto-market capitalization mainly through financial stability, counterparty-risk reduction, and institutional investability rather than through direct effects on token prices. Repeated failures of under-capitalized crypto-asset service providers, including exchanges and custodians, highlight the role of weak solvency and liquidity buffers as a key source of vulnerability in crypto markets, often leading to direct investor losses and sharp declines in market confidence [3]. Financial-economics research shows that thin capitalization increases both the likelihood and severity of intermediary distress and failure, generating negative externalities through asset-price dislocations and liquidity spillovers [1]. By requiring exchanges to hold minimum levels of loss-absorbing capital, MiCA reduces the probability that operational stress or market shocks translate into chaotic platform failures, which have historically been among the most powerful drivers of sudden crypto market-capitalization destruction. At the same time, own-funds requirements reshape the risk profile of intermediaries in ways that affect capital inflows. Banks, asset managers, pension funds, and payment institutions typically face binding counterparty-risk constraints that prevent engagement with trading venues or custodians lacking demonstrable solvency buffers. By introducing enforceable prudential capital standards, MiCA removes a key institutional barrier that previously limited large-scale participation in crypto markets, enabling more stable and longer-duration capital to access regulated exchanges. Although these requirements raise barriers to entry and may dampen short-term competitive intensity or regional liquidity, they also create a more resilient intermediary layer that supports long-run market credibility and capital retention. On this basis, the capital and own-funds framework is assigned an influence score of 8/10.

Customer Trust

User trust is strongly reinforced by own-funds requirements because they directly address fears of platform insolvency and sudden collapse. A consistent driver of distrust in crypto markets has been the failure of undercapitalized exchanges, often accompanied by frozen withdrawals, prolonged resolution processes, or permanent loss of client assets. By obliging exchanges to maintain continuous minimum levels of loss-absorbing own funds, MiCA lowers the perceived risk that operational stress or market volatility will translate into immediate insolvency. This strengthens user confidence that an exchange can withstand adverse conditions without jeopardizing customer balances. The trust effect is also amplified through signalling. For retail users, regulatory capital aligns exchanges more closely with traditional financial intermediaries, conveying institutional maturity and accountability. For institutional participants, capital adequacy is frequently a binding prerequisite, since internal counterparty-risk frameworks at banks, asset managers, and payment institutions often prohibit engagement with venues lacking enforceable solvency buffers. As a result, own-funds requirements materially expand the set of exchanges that institutions can trust for custody and execution, with positive spillovers to retail confidence. At the same time, this effect is not absolute: users rarely observe capital positions in real time, disclosures can be opaque, and capital does not prevent fraud. Nevertheless, relative to other measures, own-funds requirements most directly address the fear of irrecoverable platform collapse. On this basis, capital and own-funds requirements are assigned a signed customer-trust score of 9/10.

Security

Own-funds requirements do not alter the technical probability of a successful cyberattack against an exchange. These provisions are designed to ensure solvency and loss-absorption capacity, not to govern cyber controls such as access management, encryption, monitoring, or incident response. While higher capital can improve an exchange's ability to absorb losses after an incident, it does not reduce the likelihood of compromise beforehand. On this basis, the capital and own-funds framework is assigned a signed security score of 0/10.

Compliance Costs

The capital and own-funds requirements under MiCA impose one of the most financially binding compliance burdens on crypto exchanges because they mandate the continuous holding of balance-sheet capital that cannot be deployed for growth or operations. Unlike licensing, outsourcing, or governance obligations, which primarily generate administrative and organizational costs, capital requirements create an ongoing opportunity and funding cost. Although MiCA specifies baseline own-funds thresholds of €50,000 to €150,000, KPMG notes that CASPs must at all times hold capital equal to at least the regulatory minimum or a quarter of fixed overheads, whichever is higher, and must cover a broad range of operational risks and wind-down obligations—implying materially higher effective capital requirements for larger or more complex firms [22]. Beyond the capital itself, firms also face recurring prudential compliance costs related to capital monitoring, internal capital adequacy processes, stress testing, and regulatory reporting. Taken together, these factors justify assigning MiCA's capital and own-funds framework a compliance cost score of 9/10.

3.1.5 Segregation of Client Assets

The Segregation of Client Assets establishes rules to ensure that client crypto-assets and client funds are legally and operationally separated from the proprietary assets of exchanges. The core objective of this policy is to protect clients in the event of exchange insolvency, operational failure, or misuse of assets, preventing client holdings from being treated as part of the firm's own estate. Exchanges must implement operational, legal, and accounting arrangements that ensure continuous segregation of client assets. Client crypto-assets and funds must be held in separate accounts, clearly identifiable at all times, and exchanges are prohibited from using client assets for their own account without explicit and informed client consent. The regulation further requires exchanges to maintain accurate record-keeping systems that allow for immediate reconciliation between client entitlements and the assets actually held. In the event of insolvency, client assets must be returned to clients as a matter of priority under applicable national insolvency law. These provisions directly address one of the most severe historical weaknesses in crypto markets: the absence of enforceable legal separation between client property and exchange balance sheets. As a result, segregation of assets is one of the strongest investor-protection tools within MiCA's framework.

Market Cap

Segregation of client assets influences crypto-market capitalization primarily by reducing risk and preventing the destruction of existing capital during exchange failures. Some of

the most severe historical contractions in crypto market value have followed the failure of exchanges and custodians that commingled client assets with proprietary funds, resulting in irreversible customer losses and abrupt collapses in market confidence. Financial-crisis research shows that uncertainty about asset values and liquidity conditions can amplify run-style dynamics and systemic contagion, with small shocks disproportionately affecting liquidity and asset prices [2]. In crypto markets, the absence of segregation has repeatedly transformed firm-level distress into market-wide sell-offs through sudden withdrawal freezes, forced liquidations, and cascading loss of confidence across multiple tokens. By mandating return of client assets, MiCA weakens this transmission channel. Exchange failures under a segregation framework are less likely to trigger panic-driven exits from the market, as insolvency no longer implies total loss of customer holdings. This materially improves the survivability of capital already within the cryptocurrency market, even during periods of acute stress. Segregation requirements also play a decisive role in institutional participation. Institutions are typically barred from holding assets on platforms where client property is not legally separated from proprietary risk. By resolving this legal constraint, MiCA makes long-duration institutional custody and trading structurally feasible within the EU. At the same time, segregation rules do not stimulate new crypto demand, influence issuance, or shape speculative sentiment. On this basis, segregation of client assets is assigned a market-capitalization influence score of 7/10.

Customer Trust

Customer confidence is exceptionally strengthened by segregation because it addresses the irreversible loss of customer property through commingling, misuse, or platform insolvency. Many of the deepest trust collapses in crypto history were driven not by price volatility, but by exchanges treating client assets as proprietary resources and failing to return them once distress emerged. By legally requiring that client crypto-assets and client funds be held separately from the exchange's balance sheet, MiCA shifts the customer relationship from an unsecured claim to protection in insolvency. This legal clarity sharply increases users' confidence that their assets will not simply vanish if an exchange fails, which is central to trust formation in custodial environments. Segregation also reshapes user behavior: when customers believe misappropriation risk is structurally constrained, they are less inclined to engage in precautionary withdrawals or costly self-custody purely for fear-based reasons, and more willing to retain balances and use exchange services on an ongoing basis. While segregation does not eliminate operational disruptions, valuation losses, or delays during insolvency proceedings, it removes the structural mechanism that historically turned exchange failures into total customer losses. On this basis, segregation of client assets is assigned a signed customer-trust score of 10/10.

Security

Segregation is a legal and accounting safeguard, so it does not function as a preventive cyber control. It clarifies ownership but it does not govern wallet architecture, key management, access controls, monitoring, or incident response. Assets can be segregated in records and still be stolen if technical controls are weak. On this basis, the segregation of client assets framework is assigned a security score of 0/10.

Compliance Costs

Segregation imposes high costs on exchanges because compliance often requires changes to custody infrastructure, ledger systems, and operational processes. Achieving compliance typically requires systems that maintain accurate records of client entitlements, operate segregated wallet structures or sub-accounts, reconcile balances with on-chain or custodial positions, and prevent reuse of client assets without explicit consent. Where firms previously relied on commingled wallets or basic accounting, this can entail substantial re-engineering of wallet architecture, back-office systems, internal controls, and audit processes. Industry and policy analyses emphasize that delivering institutional-grade digital-asset custody—incorporating asset segregation, advanced security and operational controls, and robust client-asset protections—entails substantial technological and operational investment, with requirements that increasingly mirror those of traditional custody frameworks [24] [28]. Transitioning to MiCA-compliant asset segregation entails substantial one-off implementation costs for crypto exchanges, particularly where new custody providers, wallet infrastructures, or reconciliation systems must be introduced. These changes also generate ongoing operational costs, as segregation requires continuous reconciliation, independent audits, control testing, and enhanced governance and reporting processes. On this basis, segregation of client assets is assigned a compliance cost score of 7/10.

3.1.6 White Paper and Disclosure Requirements

The White Paper and Disclosure Requirements establish a comprehensive transparency framework governing the public offering and admission to trading of crypto-assets within the EU. These provisions require crypto-asset issuers, except for certain exempt categories such as small offers or fully decentralized protocols without an identifiable issuer, to publish a crypto-asset white paper before offering tokens to the public or seeking admission to trading on a trading platform. Under MiCA, issuers must prepare a mandatory white paper containing detailed information on the issuer, the offered crypto-asset, functionality, intended use, pricing and distribution, and associated technological and market risks [5]. Issuers are held legally liable for false, misleading, or incomplete disclosures, substantially raising the cost of misrepresentation relative to the pre-MiCA environment. Issuers must notify their national competent authority and publish the document in a freely accessible digital format. For asset-referenced tokens (ARTs) and e-money tokens (EMTs), significantly stricter disclosure and approval requirements apply, reflecting their higher systemic relevance. This fundamentally transforms crypto-asset issuance in the EU from an unregulated promotional environment into a legally accountable disclosure framework, closely aligned with traditional investor-protection models.

Market Cap

White-paper disclosure requirements affect crypto-market capitalization by improving information quality, deterring fraud, and shaping the credibility of token issuance rather than by mechanically moving prices. Research shows that robust disclosure reduces information asymmetry and improves price discovery, helping to deter low-quality or opportunistic issuance, while weak disclosure environments exacerbate adverse selection and market mispricing and volatility [17]. Historically, crypto markets have been characterized by extreme opacity, with many token offerings based on unverifiable claims, anonymous teams, and misleading representations of token economics, contributing to repeated episodes of

speculative over-issuance followed by mass project failure and destruction of market value. By introducing mandatory, standardized disclosures and attaching civil liability to misinformation, MiCA raises the expected cost of fraud and low-quality issuance, improving the average quality of projects entering the market. This quality filter also produces a short-term contractionary effect: higher compliance costs and legal exposure are likely to reduce the volume of speculative token launches and slow listing activity in the EU, dampening near-term market-cap growth driven by issuance quantity. Over the longer term, filtering out structurally weak projects supports capital retention and confidence. Disclosure requirements are also a precondition for institutional participation in primary crypto markets, as regulated funds, exchange-traded funds (ETFs), and structured products require reliable information to meet fiduciary and regulatory obligations. On this basis, the white-paper and disclosure framework is assigned an influence score of 6/10.

Customer Trust

Trust in exchanges is strengthened by disclosure obligations because they reduce information opacity and create legal accountability around listed tokens. A persistent source of distrust in crypto markets has been the prevalence of projects with anonymous teams, vague business models, undisclosed allocations, and misleading claims. By requiring issuers to publish clear, fair, and non-misleading white papers covering token economics, governance structures, and material risks, and by attaching civil liability to false or incomplete disclosures, MiCA lowers perceived information asymmetry and the risk of deception. Even where retail users do not read each document in detail, standardized disclosure functions as a credibility signal that projects admitted to regulated venues meet minimum transparency expectations. It also enables comparison across projects, reducing reliance on marketing narratives, which capital-markets research links to higher confidence and participation. At the same time, the trust effect remains indirect. Transparency does not prevent commercial failure or price declines even when risks are disclosed, and many users do not engage with technical documentation. On balance, the white-paper and disclosure framework is assigned a customer-trust score of 6/10.

Security

Disclosure obligations do not change an exchange's technical cyber defenses. The framework targets transparency and legal liability around token offerings, not ICT controls. A project can publish a compliant white paper while still relying on vulnerable infrastructure, and a secure exchange can still list assets where disclosure is the primary regulatory tool. On this basis, the white-paper and disclosure framework warrants a security score of 0/10.

Compliance Costs

The white-paper framework constitutes a moderate cost, driven primarily by legal drafting, technical writing, regulatory notification, and ongoing updates. For issuers, a MiCA-compliant white paper functions similarly to a mini-prospectus, with prescribed content, format, and taxonomy requirements, including eXtensible Business Reporting Language (XBRL) templates under the implementing technical standards. ESMA's MiCA taxonomy and reporting guidance confirm that specific forms, formats, and templates are mandatory for different categories of white papers, adding structured reporting work on top

of narrative drafting. Available impact-assessment evidence suggests that preparing a MiCA-compliant white paper can involve substantial costs, with estimates cited by EPICENTER indicating ranges of approximately €35,000 to €75,000 per project, depending on complexity and scope [9]. Other providers similarly market entry-level services from a few thousand euros, but realistic end-to-end legal costs for a fully compliant document commonly reach the low five-figure range once internal coordination costs are included. Beyond external legal fees, issuers also incur internal costs related to management time, technical documentation, risk analysis, environmental disclosures, and alignment of marketing communications with the published white paper. Exchanges face a smaller but non-zero cost burden: they must verify the existence and validity of issuer white papers, maintain public access links for listed assets, and in certain cases prepare their own white papers when listing tokens on their own initiative. On this basis, the white-paper and disclosure framework is assigned a compliance-cost score of 4/10.

3.1.7 Product Intervention Powers

The product intervention framework under MiCA empowers the European Securities and Markets Authority (ESMA), the EBA, and national competent authorities (NCAs) to temporarily restrict or prohibit the marketing, distribution, or sale of certain crypto-assets or crypto-asset activities where serious concerns arise in relation to investor protection, market integrity, or financial stability. In particular, NCAs are authorized to impose domestic product intervention measures. These powers are supported by investigatory and enforcement mechanisms that enable authorities to request information, conduct inspections, and ensure compliance with intervention decisions. In effect, this framework allows EU regulators to remove high-risk or harmful crypto products from the legal market entirely.

Market Cap

Product intervention powers affect market capitalization through the direct exclusion of specific products or activities from the EU market, making them among the most explicitly contractionary tools in MiCA. When a crypto-asset, trading feature, or related activity is restricted or prohibited, EU-based trading volume and liquidity for that product decline immediately, leading to a mechanical reduction in its market capitalization. Applied to crypto markets, activation of MiCA's intervention powers would generate an immediate negative shock to the market value of affected tokens or activities within the EU, particularly in segments where valuations are driven largely by retail speculative demand. While some displaced capital may migrate to other crypto-assets or to non-EU venues, prior regulatory experience indicates that substitution is incomplete, implying that a portion of market value is extinguished rather than merely relocated. Removing structurally harmful or misleading products may improve market integrity and reduce the severity of retail-driven boom–bust cycles in the long run, but this protective effect does not offset the immediate contraction in restricted segments. On this basis, the product intervention framework is assigned a market-capitalization impact score of –3/10.

Customer Trust

Intervention powers provide a modest but real trust benefit by signalling that the most harmful products can be removed when retail detriment becomes evident. The knowledge that ESMA, the EBA, and NCAs can restrict or prohibit particularly toxic crypto prod-

ucts indicates that the product universe on regulated platforms is subject to an investor-protection backstop. This mirrors experience in traditional markets, where ESMA interventions against binary options and highly leveraged CFDs were framed as necessary to restore baseline confidence in affected segments. In the crypto context, similar powers make regulated exchanges appear more actively supervised, particularly for cautious users or those previously harmed by predatory or structurally loss-making products. These powers also contribute indirectly by signalling regulatory vigilance and responsiveness. However, the trust effect is bounded. Product bans are episodic rather than a continuous feature of the customer experience, and many retail users are only vaguely aware of them. The possibility of sudden restrictions or delistings can also introduce uncertainty for some users. Because these powers do not address solvency, asset segregation, or operational reliability, they function as a secondary reassurance rather than a primary trust anchor. On this basis, the product-intervention framework is assigned a customer-trust score of 4/10.

Security

Intervention powers do not systematically improve cybersecurity. They determine which products may be marketed or sold, not whether an exchange's systems are resistant to intrusion. A platform can be fully compliant with intervention decisions and still be vulnerable to attack if its technical controls are weak. Accordingly, the product-intervention framework is assigned a signed security score of 0/10.

Compliance Costs

Compliance costs arising from product intervention are generally low because obligations are reactive and episodic. Once an intervention decision is adopted, exchanges must cease to offer, market, or sell the affected product to the relevant client segment, most commonly retail users. There is no standing requirement to build new infrastructure, raise capital, or maintain a permanent governance function solely to satisfy these powers. Costs arise when interventions occur, including updating product catalogues, trading interfaces, and marketing materials; communicating with affected clients and managing restrictions or closures; and adjusting internal policies and disclosures. On this basis, the product-intervention framework is assigned a compliance-cost score of 1/10.

3.2 Comparison policy scores

Taken together (see Table 3.1), the relative scores across DORA and MiCA policies reflect clear differences in what each framework is designed to protect and how directly it addresses the core risks of crypto markets. Measures that reshape legal rights and financial resilience at the platform level score higher than those that operate indirectly or ex post. Segregation of client assets ranks highest on customer trust because it fundamentally changes ownership outcomes in insolvency, whereas own-funds requirements score slightly lower as they reduce failure risk without guaranteeing asset return. Licensing follows because it signals oversight and accountability but does not itself ensure solvency or asset protection. In the security category, DORA scores far above all other policies because it is the only framework that directly mandates technical ICT controls and resilience testing; outsourcing controls rank below it because they mitigate supply-chain risk but remain de-

Policy	**Financial Security**	**Trust**	**Market Cap**	**Not Implement Market Cap**	**Cost**
Licensing Requirement	0	8	7	-7	7
Operational Resilience / Cybersecurity (DORA)	9	7	4	-4	8
Outsourcing Controls	6	5	4	-4	5
Capital / Own Funds	0	9	8	-8	9
Segregation of Client Assets	0	10	7	-7	7
White Paper / Disclosure	0	6	6	-6	4
Ban on Certain Products	0	4	-3	3	1

Table 3.1: Summary of policy impact scores and compliance costs

pendent on third-party behavior, while all other measures score zero as they govern legal or prudential arrangements rather than cyber defense.

Market-capitalization scores reflect stabilization rather than growth effects: capital requirements rank highest because they prevent disorderly failures and enable institutional participation, while licensing and segregation follow by improving market structure and capital retention without expanding demand. Disclosure and outsourcing score lower as they mainly improve information quality and operational robustness, and product intervention scores negatively because it removes products and liquidity from the market. Compliance-cost differences mirror this hierarchy, with balance-sheet and technology-intensive obligations ranking highest, governance and operational reforms in the middle, and episodic intervention measures at the bottom. Overall, policies score higher than others when they directly and structurally reduce the risks that historically caused the greatest market harm, and lower when their effects are indirect, contingent, or narrowly targeted.

Chapter 4

Implementation

In this model, the regulatory policies are treated as optional strategic choices for analytical purposes, even though in practice they are mandatory for any exchange seeking to operate legally in the European cryptocurrency market. This abstraction is deliberate: the objective is not to replicate regulatory compliance in real time, but to examine how different types of exchanges perform under market conditions when subject to different policy configurations. All exchanges in the model are initially identical. They begin with the same level of trading volume, capital, and baseline market conditions, and they are not differentiated by size, liquidity, or customer base. The only dimension along which exchanges may differ is their strategy profile, which represents the set of regulatory and operational policies they adopt. Multiple exchanges may select the same strategy at the same time. This design choice isolates the effects of policy selection from confounding factors such as scale, market power, or incumbency advantages. The purpose of the model is to observe how market outcomes evolve as exchanges adopt different policy strategies and interact with one another, and to analyse what these dynamics imply for the future structure of cryptocurrency markets and for exchange behaviour in a long-run, strategic sense. The model does not attempt to simulate real-world political processes, macroeconomic shocks, or exogenous economic events that also influence crypto markets. Instead, it focuses narrowly on exchange-level actions and the market consequences that follow from those actions. The framework is implemented as a multi-player game-theoretical model with five exchange players, chosen for analytical convenience and to correspond to five distinct strategy types. Attackers and regulatory authorities are not modelled as active strategic players. Instead, they appear as passive background elements represented through exogenous policy constraints and breach probabilities. For example, attackers do not make strategic choices within the game, but are reflected in the likelihood of security breaches conditional on the policies adopted by exchanges.

4.1 Parameters and functions

This section introduces the parameters and functional relationships used to model interactions between cryptocurrency exchanges. The framework captures how market size, security investment, trust, and regulatory compliance determine trading activity, breach risk, and profitability of an exchange.

- M : Total Market capitalization of cryptocurrency.
- x : Security score of an exchange.

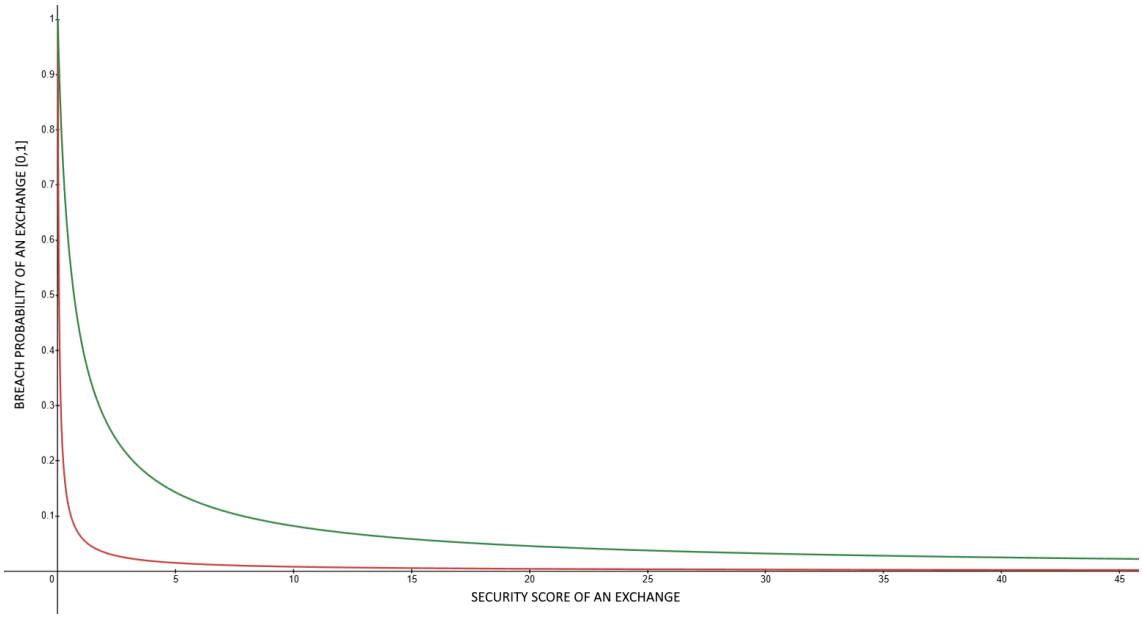

Figure 4.1: Breach Probability Function with high and low market capitalization

- C : Compliance or regulatory cost.
- μ : Revenue margin per unit of volume (scalar).
- ℓ : Loss per unit of volume in the event of a breach (scalar).
- α : Scale parameter linking market capitalization to volume (scalar).
- β : Elasticity of volume with respect to market capitalization (scalar).
- δ : Trust–security coupling parameter.
- t : Trust score of an exchange.
- κ : Effectiveness of security investment in reducing breach risk.

$$\tau = \frac{\delta x t^f}{\sum_{i=1}^{4} t_i^f x_i + \delta x t^f} \tag{4.1}$$

where τ denotes the trust score relative to other exchanges.

$$\lambda_i = \frac{1}{1 + \left(\frac{x_i}{V}\right)^{\kappa}} \tag{4.2}$$

where λ_i denotes the probability of a successful security breach for exchange i.

$$Q = \alpha\, M^{\beta} \tag{4.3}$$

where Q denotes the trading volume of the exchange as a function of its market capitalisation M.

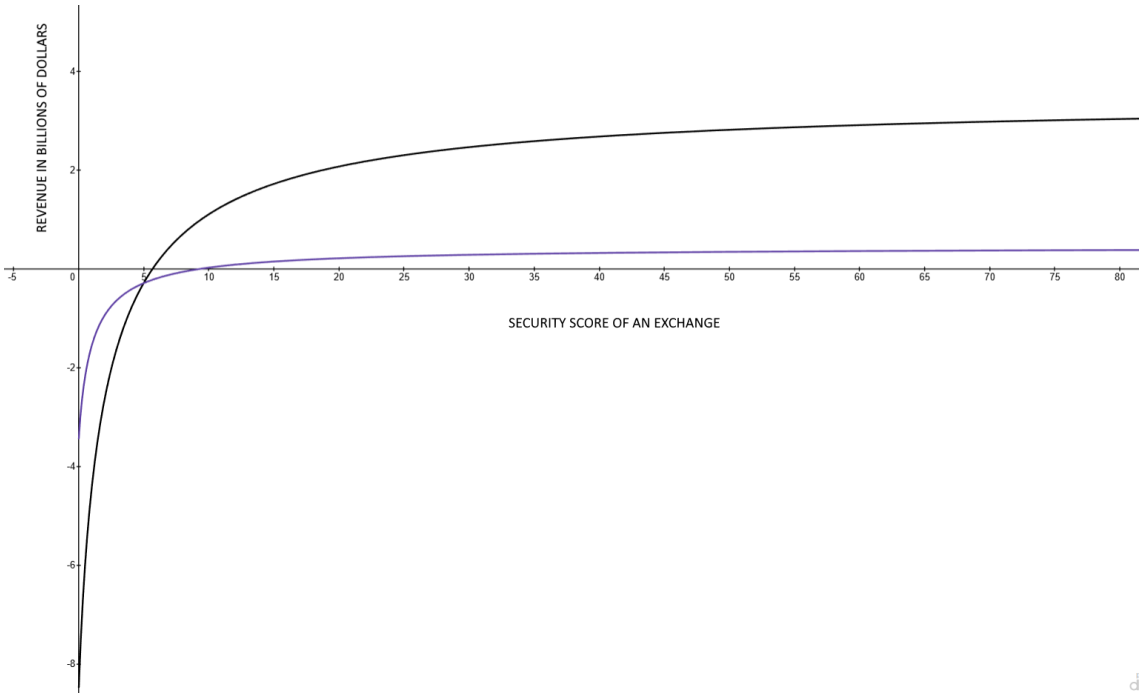

Figure 4.2: Utility Function of an exchange with high — and low — market capitalization

$$\Pi = \tau\, Q\, \mu \;-\; Q\, \ell\, \lambda \;-\; C \tag{4.4}$$

where Π denotes the expected profit of the exchange.

4.2 Discussion

As illustrated in Figure 4.1, higher market capitalization is associated with a greater likelihood of security breaches, consistent with an increase in the frequency and intensity of attacks as the market grows. In such environments, exchanges with lower levels of security investment face a substantially higher risk of breach. By contrast, when market capitalization is low and the ecosystem is less mature, relatively modest security investment may be sufficient because attacks are less prevalent; however, this does not imply that security investment is unnecessary at earlier stages of market development. The figures also illustrate diminishing returns to security investment: beyond a certain point, additional investment yields progressively smaller reductions in breach risk, as discussed in Section 2.1.

The relationship between Q and M is modeled using a power law, reflecting empirically observed scaling behavior between market size and trading activity in financial markets [6].

In Figure 4.2, higher market-capitalization environments require stronger security, since higher breach incidence implies larger expected losses. It is important to note that this figure assumes a constant trust score across exchanges; therefore, it isolates the effect of security investment on utility. The strategy profile in which all exchanges implement all policies is used for this illustration. When strong security measures are applied we can see substantial returns for the exchange.

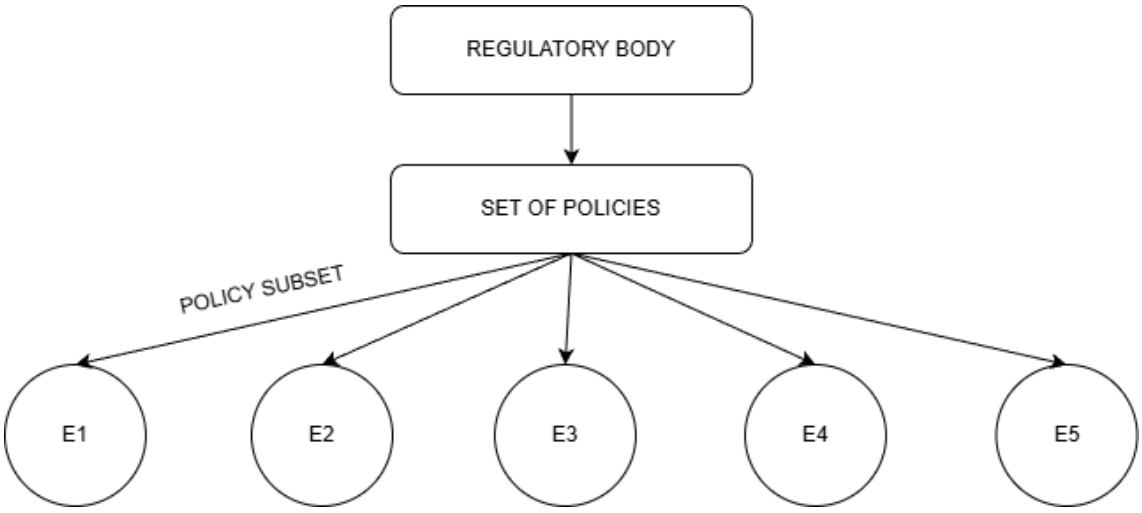

Figure 4.3: Sequential Game

4.3 Simulation

The simulation is structured as a repeated game played over 120 discrete rounds, with each round representing one month. Collectively, the simulation therefore spans a ten-year period, allowing the model to capture medium to long-term market dynamics rather than short-term fluctuations. This extended timeframe is chosen to observe how the cumulative effects of different policy and strategy choices unfold over time in the cryptocurrency exchange ecosystem. Payoffs 2.2.3 in the model are expressed in billions of dollars for interpretability, but they are not intended to correspond directly to real-world market capitalization or firm valuation. As with other model variables such as compliance costs, which are represented as relative scores rather than actual expenditures. It directly affects each exchange's outcomes within the simulation and allows for meaningful comparison across strategies. At the start of the game, each exchange selects a single strategy profile. A strategy profile consists of a predefined set of regulatory policies drawn from those made available by the regulatory framework (see Figure 4.3). Once chosen, the strategy is fixed: exchanges do not revise, adapt, or switch strategies during the simulation. Each exchange therefore plays all 120 rounds under the same policy configuration. This design isolates the long-run consequences of initial strategic choices and avoids conflating policy effects with dynamic optimization or learning behavior. The repeated-round structure is central to the model's purpose. By observing outcomes over 120 months, the simulation captures how market conditions evolve endogenously in response to exchanges' policy choices, including the accumulation of trust effects, security outcomes, cost burdens, and capital dynamics. The model is deliberately narrow in scope, focusing on the strategic behavior of exchanges and the market consequences that arise from those behaviors, rather than attempting to replicate real-world market prices or macroeconomic shocks. The simulation code is available in the associated GitHub repository: `https://github.com/sourloean/Securing-Cryptocurrency-Exchanges-Optimal-Policy-Design.git`.

4.3.1 Full Policy Adoption

Under this strategy, an exchange adopts the complete set of regulatory and operational policies recommended by the regulatory framework without modification or selective exclusion. All policies included in the model are implemented simultaneously, regardless of

their individual effects on security, customer trust, market capitalization, or compliance costs. This strategy represents a maximally compliant and risk-averse posture, serving as a benchmark against which more selective strategies can be evaluated. It approximates the behavior of exchanges that prioritize regulatory certainty, institutional credibility, and long-term market legitimacy over short-term cost or flexibility considerations.

4.3.2 Security-Focused Policy Adoption

In this strategy, an exchange adopts only those policies that demonstrably reduce the likelihood or expected severity of security breaches. Policy selection is based exclusively on the security dimension of the model: only policies that receive a strictly positive integer score in the security category are implemented. Policies that are neutral or negative with respect to breach risk are excluded, regardless of their effects on customer trust, market capitalization, or compliance costs. This strategy captures the behavior of exchanges that prioritize technical resilience and breach prevention as their primary objective, viewing cybersecurity risk as the dominant threat to survival and competitiveness.

4.3.3 Trust-Focused Policy Adoption

Under this strategy, the exchange selects policies based on their contribution to customer trust. All policies in the model are compared along the customer-trust dimension, and the arithmetic mean trust score across policies is calculated. The exchange then adopts all policies whose trust score is greater than or equal to this mean value. Policies with below-average trust effects are excluded, even if they perform well in other dimensions. This strategy reflects an exchange that competes primarily on perceived safety, legitimacy, and reputational strength, aiming to maximise users' willingness to deposit and retain assets on the platform. Strategy 4: Market-Capitalization-Oriented Policy Adoption In this strategy, the exchange prioritizes policies that are expected to have a relatively strong positive effect on market capitalization. As with the trust-based strategy, all policies are ranked according to their market-capitalization scores, and the mean score is calculated. The exchange adopts all policies with scores greater than or equal to the mean, while excluding those with below-average effects. This strategy represents an exchange that focuses on capital formation, liquidity retention, and long-term valuation dynamics, even if such policies entail higher compliance costs or weaker performance in other dimensions.

4.3.4 Cost Minimizing Policy Adoption

This strategy is driven by compliance-cost considerations. The exchange evaluates all policies according to their compliance-cost scores and calculates the mean cost score across the policy set. It then adopts only those policies whose cost score is less than or equal to the mean, excluding policies that impose above-average financial or operational burdens. This strategy captures the behaviour of cost-sensitive exchanges, particularly those with limited resources or aggressive growth objectives, which seek to minimise fixed regulatory overhead while still implementing a non-trivial subset of the available policy options.

4.4 Finding Nash Equilibrium and Social Optimum

To identify Nash equilibria 2.2.5, we evaluated all possible strategy profiles in the game. With five players and five available strategies, this yields 3,125 distinct strategy combinations. For each combination, the simulation was run for 120 rounds, and unilateral deviations were tested by allowing each exchange, in turn, to switch to an alternative strategy while all others held their strategies fixed.

Using this procedure, two pure-strategy Nash equilibria emerged. The first occurs when all exchanges adopt the strategy of implementing all regulatory policies. In this equilibrium, all players earn strictly positive payoffs. The second equilibrium arises when all exchanges select the low-cost strategy. Although no individual exchange has an incentive to deviate in this case, the resulting payoffs are negative for all players, indicating a coordination failure in which collectively inferior outcomes persist despite strategic stability.

Social optimality 2.2.4 is defined as the strategy profile that maximizes the sum of payoffs across all exchanges. Among all 3,125 possible combinations, the profile in which all exchanges implement the full set of regulatory policies yields the highest aggregate payoff. Consequently, the socially optimal outcome coincides with the Nash equilibrium characterized by universal policy adoption and positive individual profits.

To quantify the efficiency of equilibrium outcomes relative to the social optimum, we computed the PoS 2.2.7 and the PoA 2.2.6. Because some equilibria yield negative aggregate welfare, an affine offset was applied uniformly to total welfare values to ensure positivity without altering equilibrium rankings or strategic incentives.

The resulting PoS equals one, indicating that at least one Nash equilibrium achieves the socially optimal outcome. In contrast, the PoA exceeds one, reflecting a substantial efficiency loss associated with the worst Nash equilibrium. This divergence highlights that while efficient coordination is possible in the model, decentralized competition can also sustain a stable but highly inefficient equilibrium. The coexistence of these equilibria underscores the fragility of market outcomes in the presence of trust and compliance externalities and provides a clear rationale for regulatory intervention aimed at eliminating welfare-reducing equilibria and coordinating agents on the socially optimal outcome.

4.5 Results

The simulation results demonstrate that coordinated adoption of the full regulatory policy set plays a decisive role in shaping exchange behavior and overall market outcomes within the model. When all exchanges implement the complete set of policies, the game converges to a stable equilibrium characterized by positive payoffs for all participants and the maximization of aggregate welfare. In contrast, when exchanges coordinate on alternative strategies most notably the low-cost policy profile the resulting equilibrium, although stable, generates negative payoffs for all players, revealing a clear coordination failure. These outcomes indicate that selectively minimizing compliance costs can be collectively detrimental, even when individually rational. Overall, the findings suggest that collective adherence to robust regulatory standards is critical for achieving market stability, sustaining exchange performance, and preventing welfare-reducing equilibria in the cryptocurrency ecosystem.

Chapter 5

Conclusion

In this thesis, we examined the strategic interaction of cryptocurrency exchanges operating under a unified regulatory framework, using a game-theoretical model to analyze how different policy adoption strategies shape market outcomes. By abstracting regulatory requirements into strategic choices, the model allows for a systematic evaluation of how exchange behavior responds to varying levels of regulatory commitment. While the framework necessarily relies on simplifying assumptions, it provides a structured approach for assessing the long-run implications of coordinated versus selective policy adoption in crypto markets. The central finding of the analysis is that coordinated full policy adoption dominates all alternative strategy profiles in both strategic and welfare terms. When all exchanges implement the complete set of regulatory policies, the system converges to a Nash equilibrium in which no exchange has an incentive to deviate and all participants achieve positive payoffs. Importantly, this equilibrium also coincides with the social optimum, defined as the strategy profile that maximizes the aggregate payoff across all exchanges. This demonstrates that robust regulatory compliance does not merely redistribute value or impose net welfare losses, but can instead operate as a coordination mechanism that aligns individual incentives with collective outcomes.

By contrast, the model also identifies a second Nash equilibrium characterized by universal adoption of the low-cost strategy. Although strategically stable, this equilibrium yields negative payoffs for all players, illustrating a coordination failure in which individually rational cost-minimization leads to collectively inferior outcomes. This result highlights how short-term cost avoidance, when pursued uniformly, can undermine both market performance and exchange profitability. Taken together, these findings suggest that the long-run health and stability of cryptocurrency markets depend critically on coordinated adherence to strong regulatory standards. Policies that enhance operational resilience, asset protection, solvency, and trust reinforce one another over time, generating outcomes that are individually rational, collectively efficient, and welfare-maximizing. Although the model abstracts from political, macroeconomic, and behavioral complexities, it offers a clear conceptual insight: when exchanges internalize the long-term benefits of regulation and act collectively, regulatory compliance emerges not as a constraint on market performance, but as a foundation for stability, credibility, and sustainable growth.

5.1 Limitations

This thesis is subject to several important limitations that should be acknowledged when interpreting the results. First, the overall structure of the model is constrained by the

novelty of the regulatory frameworks under study. MiCA and DORA have only recently entered into force, which means that there is limited empirical evidence, historical data, and peer-reviewed research on their real-world economic effects. As a result, many of the numerical inputs used in the model such as policy impact scores, compliance costs, and market-capitalization effects are necessarily based on approximations, secondary sources, and informed judgment rather than precise, observable figures. The scoring framework therefore reflects reasoned estimation of relative effects rather than exact measurements.

Second, the mapping of policies to dimensions such as security, customer trust, market capitalization, and compliance cost involves subjective assumptions. Although these assumptions are grounded in regulatory texts, industry reports, and financial-economics literature, the true magnitude and interaction of these effects in practice may differ. In reality, the influence of a given policy may vary significantly across exchanges depending on factors such as size, technological maturity, geographic footprint, business model, and client composition. The model abstracts from these differences by treating all exchanges as initially identical, which enhances analytical clarity but limits realism.

Third, the functional form of the model and the payoff structure are stylized representations of economic performance rather than direct proxies for real-world revenue, profitability, or market capitalization. While the model is designed to approximate real-world incentives as closely as possible, it cannot capture the full complexity of exchange behavior. Numerous factors that affect exchange outcomes such as macroeconomic conditions, interest rates, global liquidity cycles, geopolitical events, competitive innovation, and user sentiment are deliberately excluded. Regulatory enforcement intensity, reputational shocks, legal disputes, and technological breakthroughs are also held constant or omitted, even though they play a significant role in shaping real-market dynamics. Fourth, the strategic environment is intentionally simplified. Exchanges are assumed to commit to a single strategy for the entire ten-year simulation period, with no learning, adaptation, or strategy switching. In practice, exchanges continuously adjust their compliance posture, security investments, and business models in response to competitors, regulators, and market conditions. Similarly, attackers and regulators are modelled as passive background forces represented through breach probabilities and policy sets, rather than as strategic agents who actively respond to exchange behavior. This limits the model's ability to capture dynamic regulatory enforcement, adversarial adaptation, or strategic lobbying.

Finally, the scope of the analysis is deliberately narrow. The thesis focuses exclusively on cryptocurrency exchanges, whereas the broader crypto ecosystem includes issuers, wallet providers, custodians, miners or validators, decentralized protocols, users, and institutional intermediaries, all of which interact in complex ways. Taken together, these limitations mean that the results should be interpreted as illustrative rather than predictive. The model does not claim to forecast actual market outcomes or regulatory impacts with precision. Instead, it provides a conceptual and analytical framework for understanding how regulatory policies can shape strategic incentives, coordination outcomes, and long-run market stability in cryptocurrency exchange markets.

A further limitation relates to the specification of the model's functional forms and parameters. The functions used to link regulatory policies to trust, security risk, trading volume, and payoffs were constructed to generate economically intuitive behavior and to highlight specific strategic outcomes of interest. As such, the choice of functional forms and parameter values is necessarily stylized and, to some extent, arbitrary. Alternative functional specifications or parameter calibrations could produce materially different quantitative results and potentially alter equilibrium outcomes. The structure of the security–investment

and breach–probability relationships is informed by the economic intuition developed in the Gordon–Loeb framework, but it is adapted to the context of cryptocurrency exchanges and regulatory compliance.

5.2 Future Work

This thesis represents an initial step toward understanding the strategic interaction between cryptocurrency exchanges and regulatory policy within a formal game-theoretical framework. As such, the model only scratches the surface of a much richer and more complex economic environment, leaving substantial scope for future research. One natural extension would be to incorporate additional economic conditions and exogenous variables in order to increase realism. Future models could explicitly account for macroeconomic factors such as interest rates, liquidity cycles, market volatility, and global risk sentiment, all of which materially influence trading volumes, revenues, and exchange behavior. Introducing stochastic shocks or framework changes would allow the analysis to move beyond a deterministic setting and better reflect the uncertainty inherent in crypto markets.

The strategic space of the model could also be expanded. In this thesis, exchanges commit to fixed strategies over the entire simulation period. Future work could allow for dynamic strategy adjustment, learning, and adaptation, enabling exchanges to respond endogenously to competitors' actions, regulatory outcomes, or realized security incidents. More sophisticated strategies such as mixed strategies, conditional compliance, or phased policy adoption—could provide deeper insight into transitional dynamics and path dependence in regulated crypto markets.

Finally, future work could improve the statistical grounding of the model. As MiCA, DORA, and comparable regulatory frameworks mature, more empirical data will become available on compliance costs, breach incidence, market reactions, and institutional participation. This would allow policy impact scores, payoff functions, and breach probabilities to be calibrated using observed data rather than informed estimation. Such calibration would strengthen the quantitative credibility of the model and allow for more robust sensitivity analysis and empirical validation.

Overall, extending the model along these dimensions would move the analysis closer to real-world complexity and enhance its usefulness for policymakers, exchanges, and researchers seeking to understand the long-run implications of regulatory design in cryptocurrency markets.

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Appendix

A.1 Declaration on the Use of Generative Artificial Intelligence

☐ **I have not used** any generative AI tools.

✓ **I have used generative AI tools.** I have verified the content generated by AI, ensured the accuracy of the outputs, and properly indicated each instance of use in the table below.

Usage type	**Name of Generative AI Tool(s)**	**Affected Sections** (chapter, page number, reference)	**Estimate Proportion of Use** (per usage type)
Literature Review	ChatGPT		
Brief Summary of the Prompt	AI was used to assist in identifying relevant literature to my topic throughout the chapters.		
Program Code Generation	ChatGPT	Chapter 4	
Brief Summary of the Prompt	To save time when doing the simulation python code was written with AI help		
Generating New Ideas or Solution Proposals	ChatGPT	Chapter 3	
Brief Summary of the Prompt	For scoring the policies I needed some opinion on what score would suit the best		
Creating Text Blocks	ChatGPT		
Brief Summary of the Prompt	AI was used for rewording sentences throughout the chapters		
Aggregated Percentage Value (for the core part of the task):			10%

Brief Textual Justification of the Aggregated Value:
The functions, main ideas, and core model are entirely my own or they have been properly referenced. Generative AI tools were used solely to assist with wording and to help identify relevant literature, as this topic is not widely discussed in existing research.